

The CURE-NURSE merger in production and perception

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Introduction

Introduction

- We investigate the CURE-NURSE merger in the eastern United States
- We find that:
 - merger in perception (MiP) is advancing while merger in production (MiPr) is obscured by variation
 - perception correlates with production in a non-straightforward way

Subsets of CURE

Lexical set	Description	Examples
PURE	CURE following a non-coronal and yod	<i>cure, pure, demure</i>
SURE	CURE following a coronal and (historic) yod	<i>sure, endure, manure</i>
TOUR	CURE elsewhere	<i>tour, poor, moor</i>

Past reports of merger

CURE-FORCE merger	• historically in the South (Kurath & McDavid 1961; Thomas 2001, 2008) • New York City and the Mid-Atlantic (Labov, Ash & Boberg 2006) • Western Pennsylvania (Thomas 2001)
PURE-NURSE (and SURE-NURSE?) merger	• west of the Appalachians (Thomas 2001; see also Guenter 2000) • younger speakers in the South (Thomas 2008)

Methods

Overview

- Data collected in early 2024
- 737 participants recruited on Prolific
 - Self-reported native English speakers
 - Primarily from NY, NJ, PA, DE or MD
- 3-part survey hosted on phonic.ai
 - Word list
 - Rhyme judgment task (Johnson 2010)
 - Demographic survey

Word list

- CURE is infrequent to be usefully studied using speech corpora
 - Excluding *your/you're*, average of 2.59 tokens per speaker in the Corpus of New York City English (Tortora et al. 2025)
- Word list contained 50 words
 - PURE: *cure, impure, obscure, pure, secure*
 - SURE: *brochure, endure, entrepreneur, immature, lure, manure, sure*
 - TOUR: *contour, detour, gourmet, poor, tour*
- FLEECE, LOT and THOUGHT included as reference vowels

Rhyme judgment task

- A minimal pair task was not feasible due to the lack of minimal pairs
- 55 word pairs:
 - 5 pairs each from 8 target mergers
 - 10 distractor pairs
 - 5 rhymes (e.g., *beak-meek*)
 - 5 non-rhymes (e.g., *wet-lap*)
 - 5 /uw/-/Cjuw/ pairs (e.g., *boot-mute*)
- Presented in writing

Demographic questions

- Open response:
 - “Year of birth”
 - “Hometown (where you spent the most of your childhood)”
 - “Race/ethnicity”
 - “Gender”
 - “What language(s) did you speak at home as a child?”
 - “Highest level of education completed”

Sample

Rhyme

- 645 participants
- 362 women; 268 men; 15 other/unknown
- 394 white; 84 Black; 68 AAPI; 57 Hispanic/Latine; 42 other/unknown
- 194 NYC; 110 Mid-Atlantic; 42 Inland North; 42 Western PA; 10 Western New England; 247 other/unknown

Production

- 554 participants
- 296 women; 212 men; 46 other/unknown
- 324 white; 63 Black; 51 AAPI; 47 Hispanic/Latine; 69 other/unknown
- 150 NYC; 90 Mid-Atlantic; 37 Inland North; 37 Western PA; 9 Western New England; 231 other/unknown

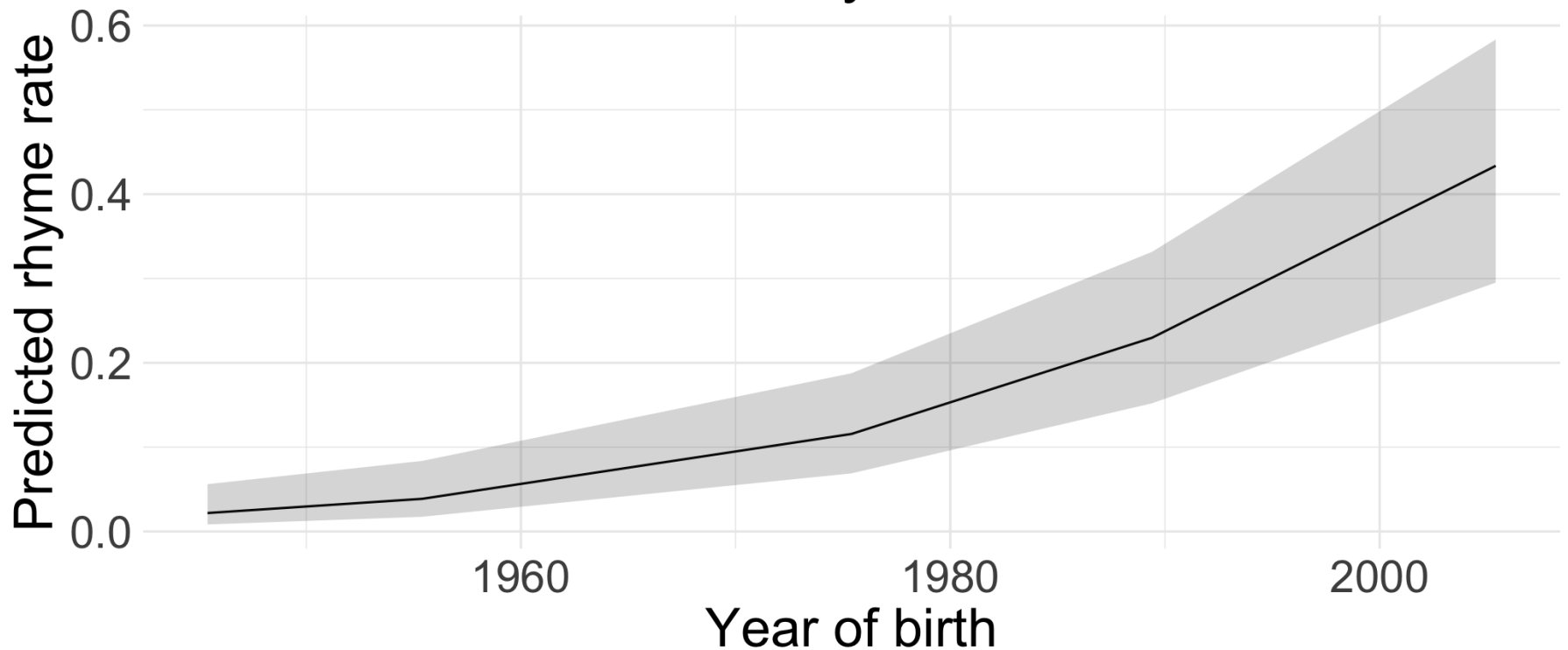
Data processing for production

- Extracted formants using DARLA's semiautomated pipeline (Reddy & Stanford 2015)
- Removed outliers using modified Mahalanobis distance (Leys et al. 2018)
- Normalized formants using ΔF (Johnson 2020, as implemented in Stanley 2023)
- Calculated by-token Euclidean distance from the by-speaker mean for NURSE

Results

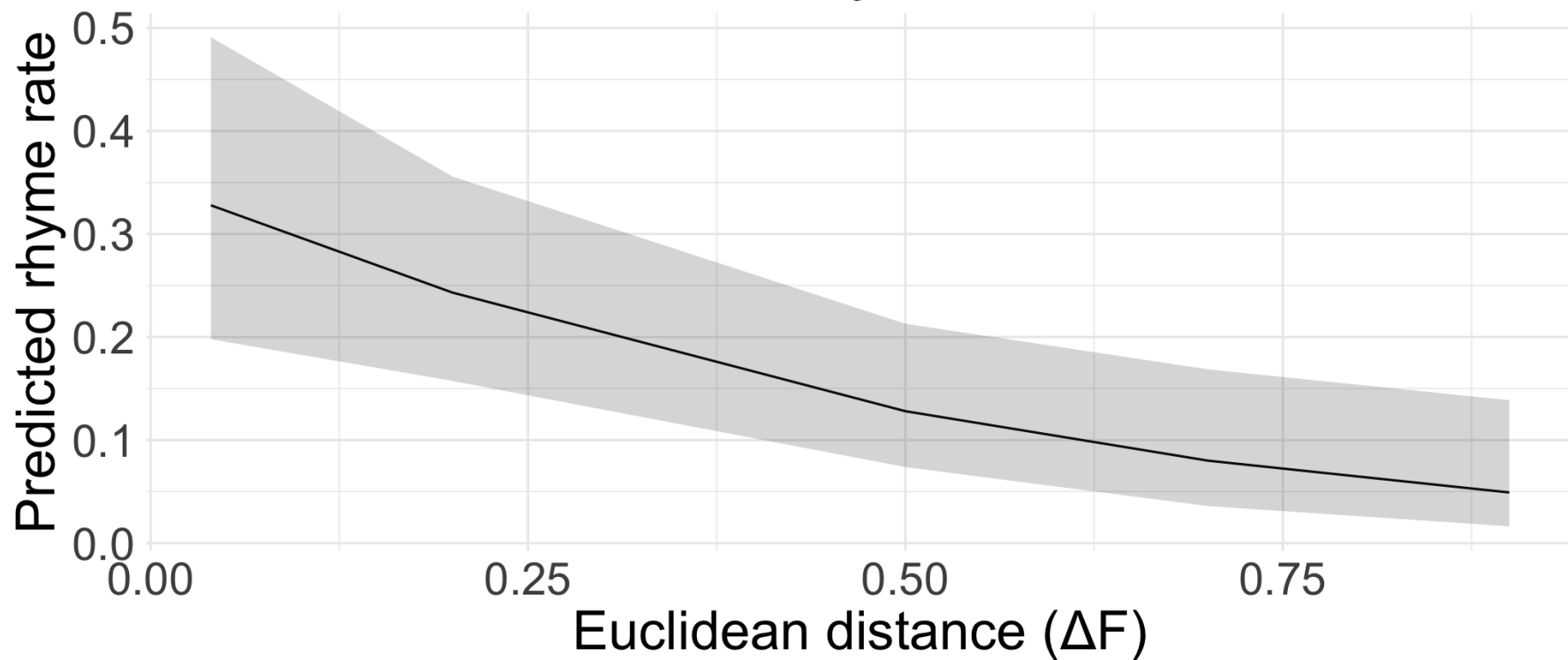
PURE/SURE-NURSE MiP is advancing

Estimated effect of age on
PURE/SURE-NURSE rhyme rate



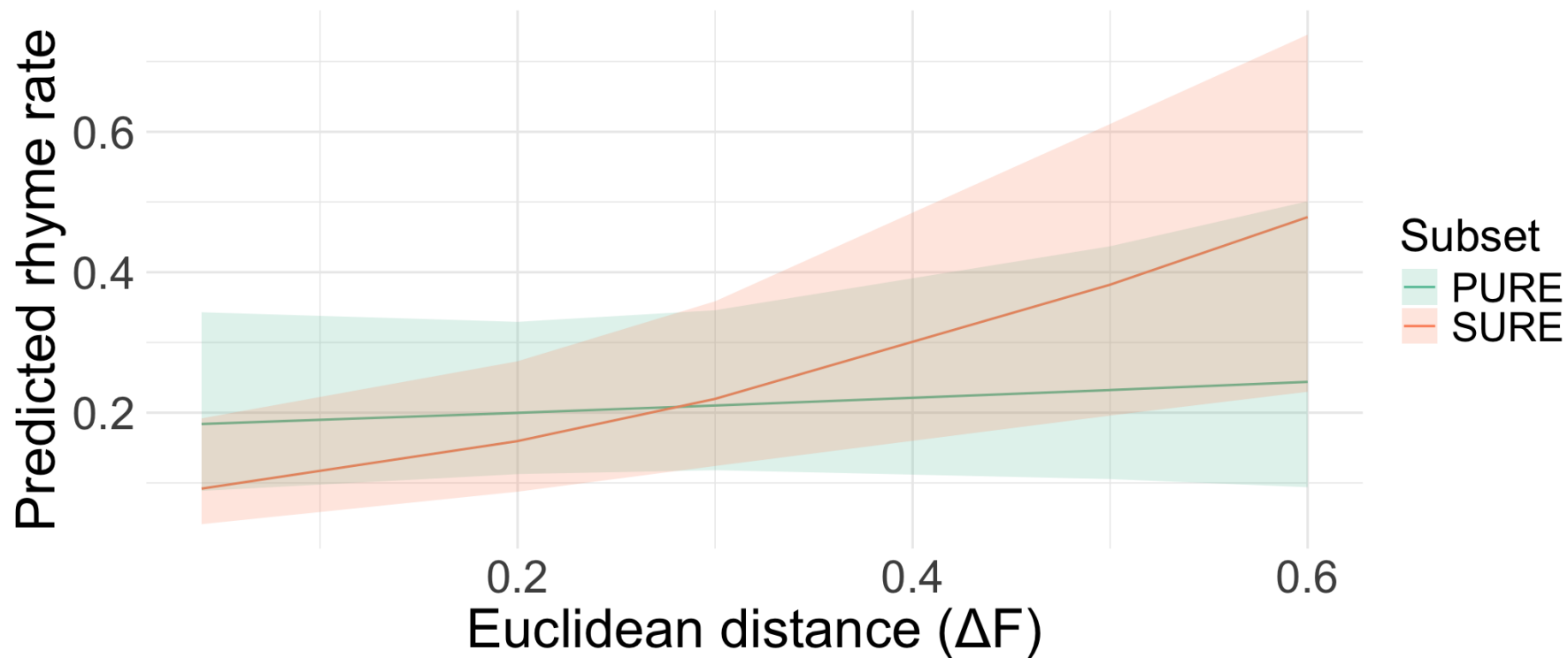
Greater phonetic distance disfavors MiP...

Estimated effect of SURE-NURSE distance on PURE/SURE-NURSE rhyme rate

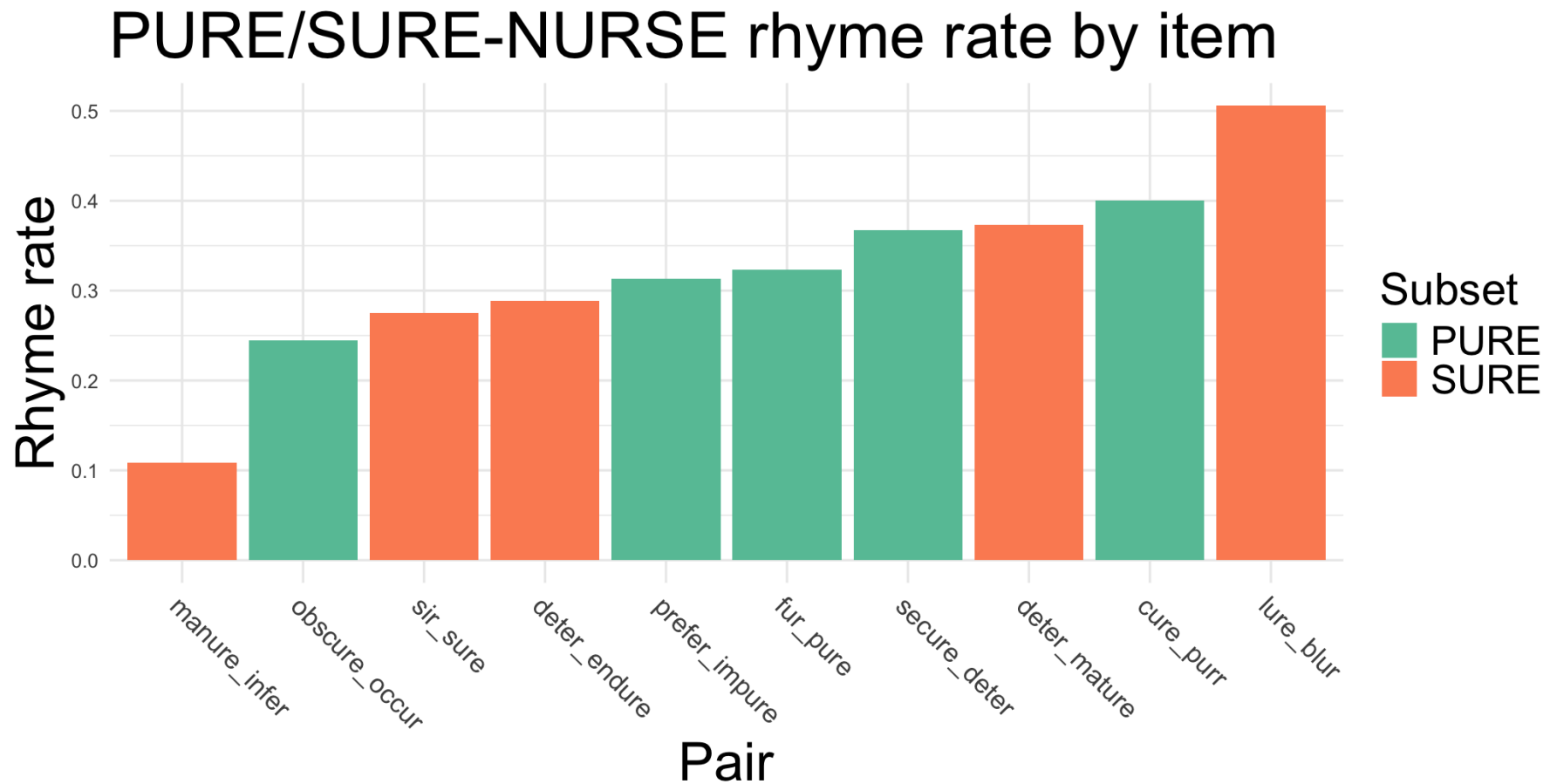


Except when it doesn't

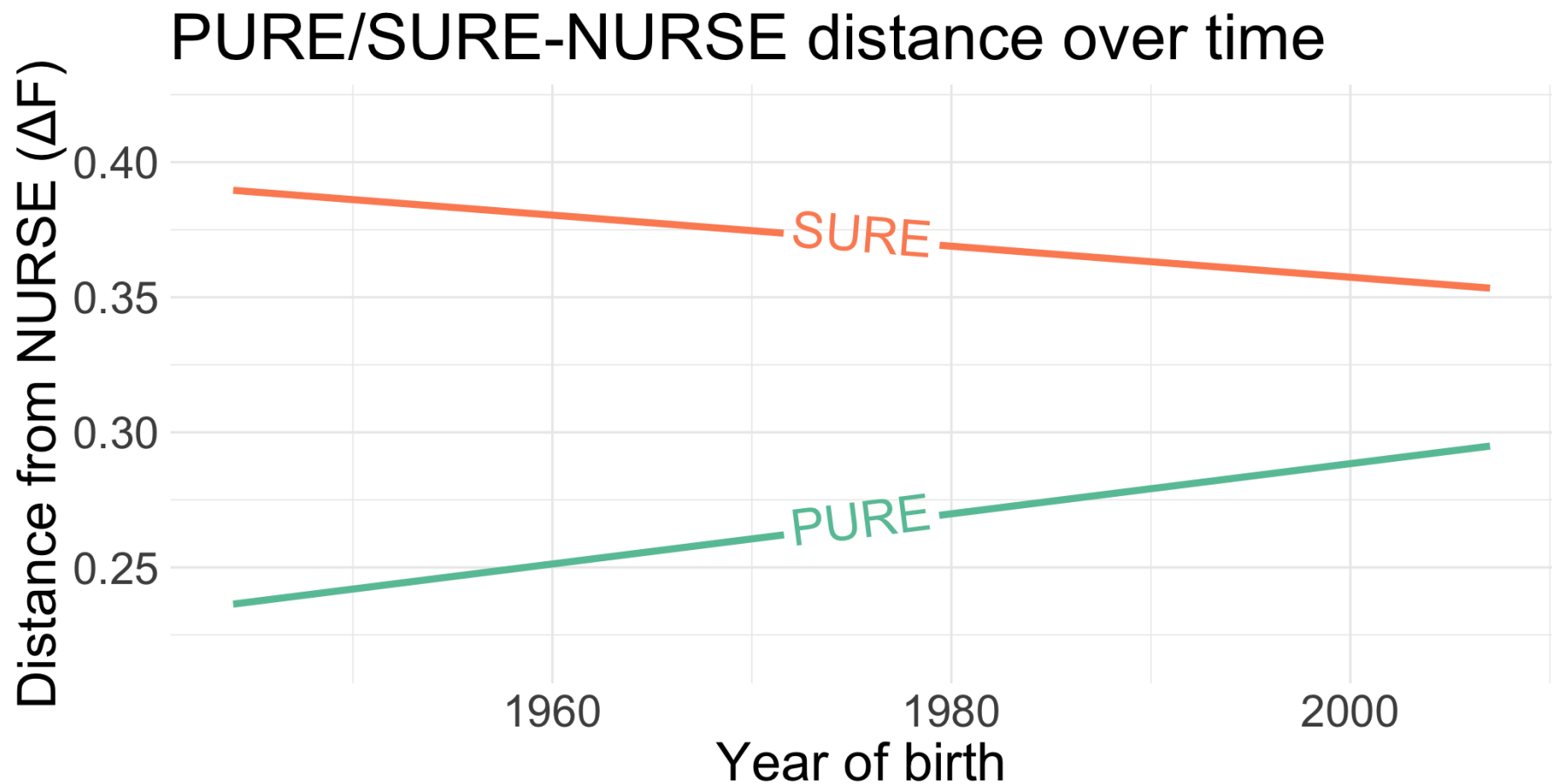
Estimated effect of PURE-NURSE distance on
PURE/SURE-NURSE rhyme rate



Words aren't all the same

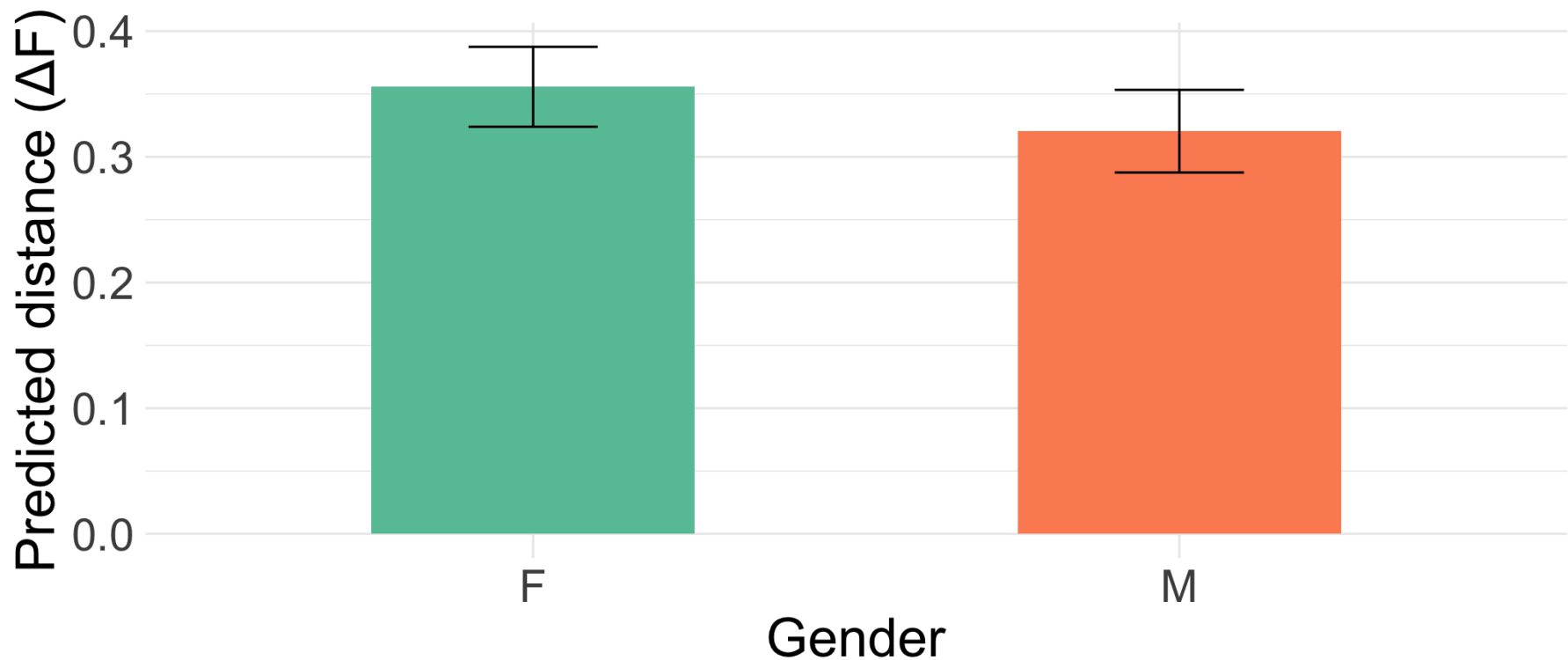


Age is not a significant predictor of PURE/SURE-NURSE distance



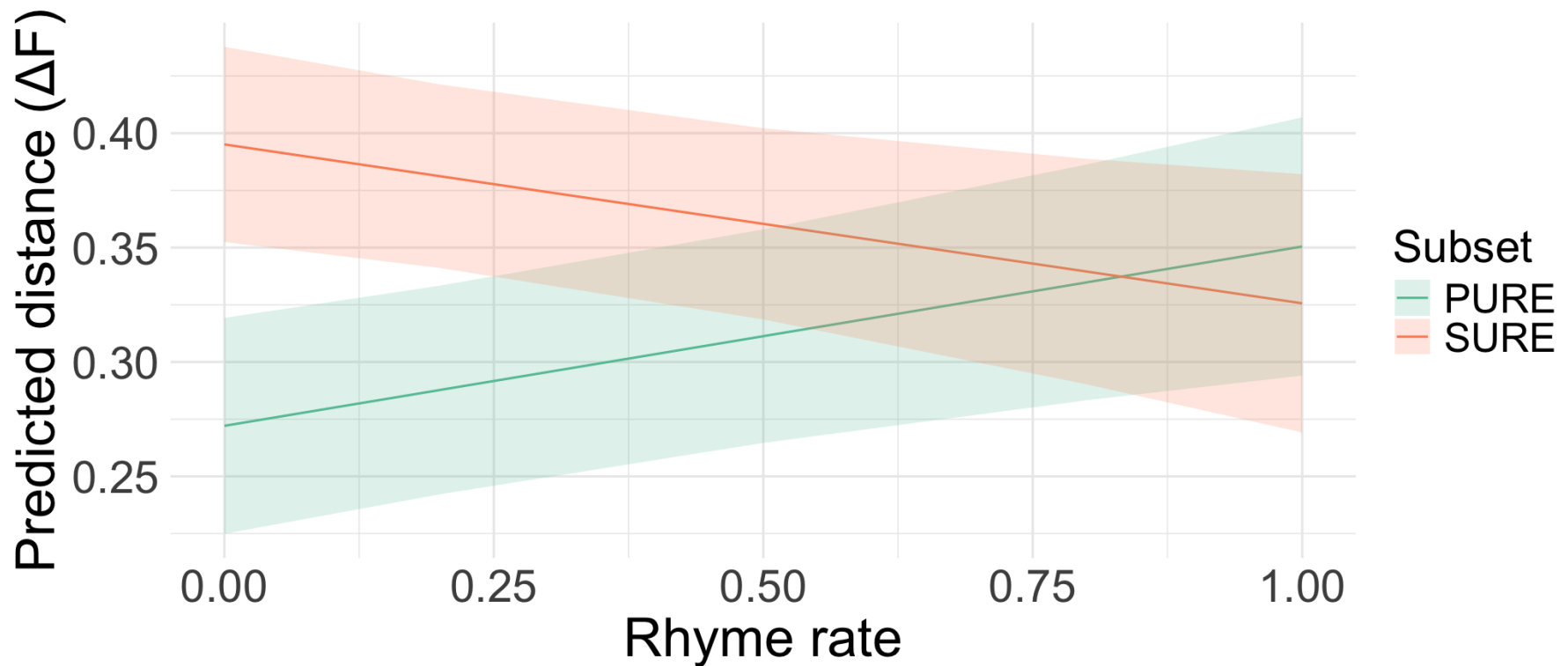
Men have PURE/SURE closer to NURSE than women

Estimated effect of gender on
PURE/SURE-NURSE distance



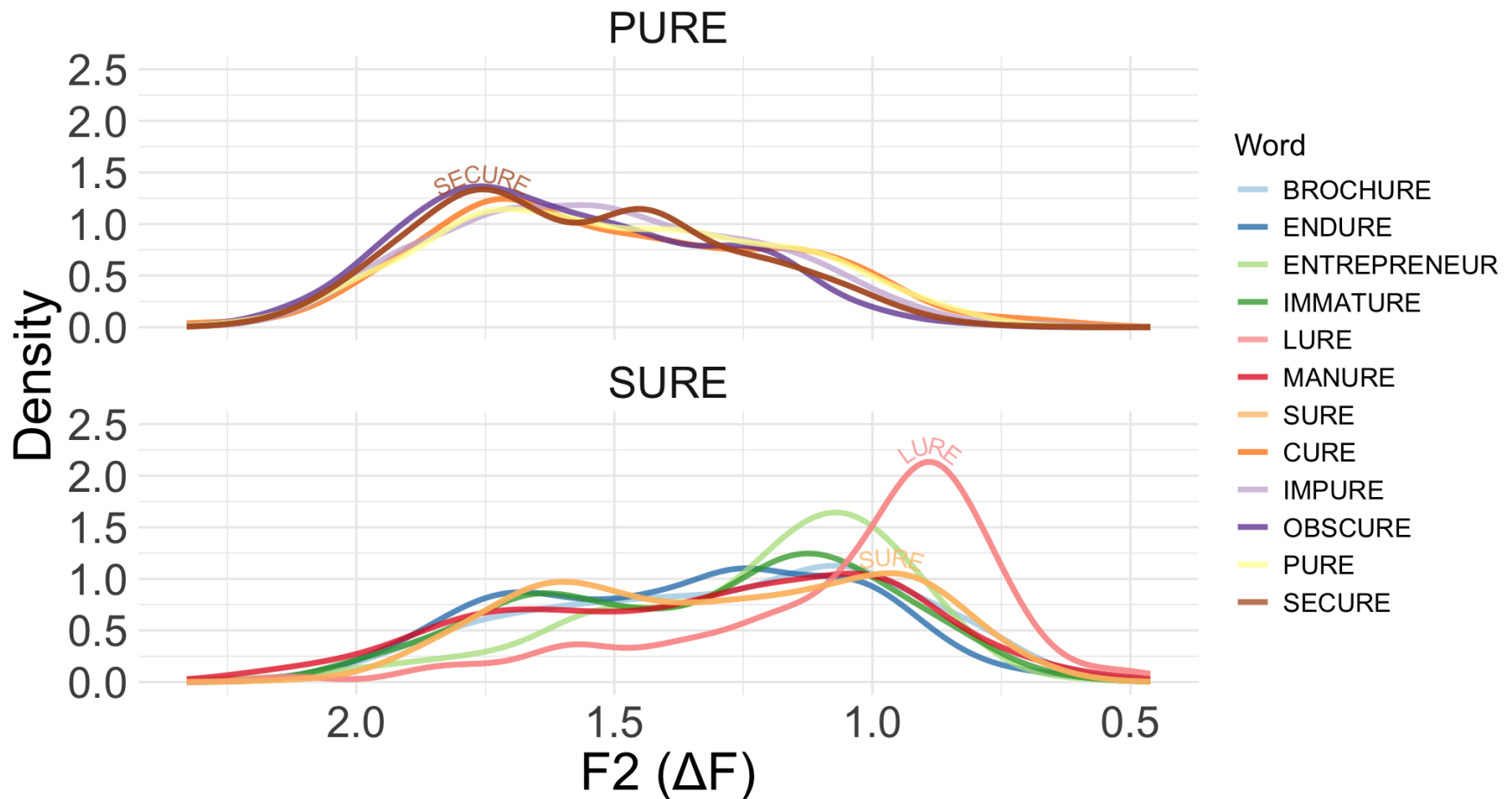
MiP predicts MiPr for SURE (but PURE is being weird again)

Estimated effect of SURE-NURSE rhyme rate on PURE/SURE-NURSE distance



Words aren't all the same (again)

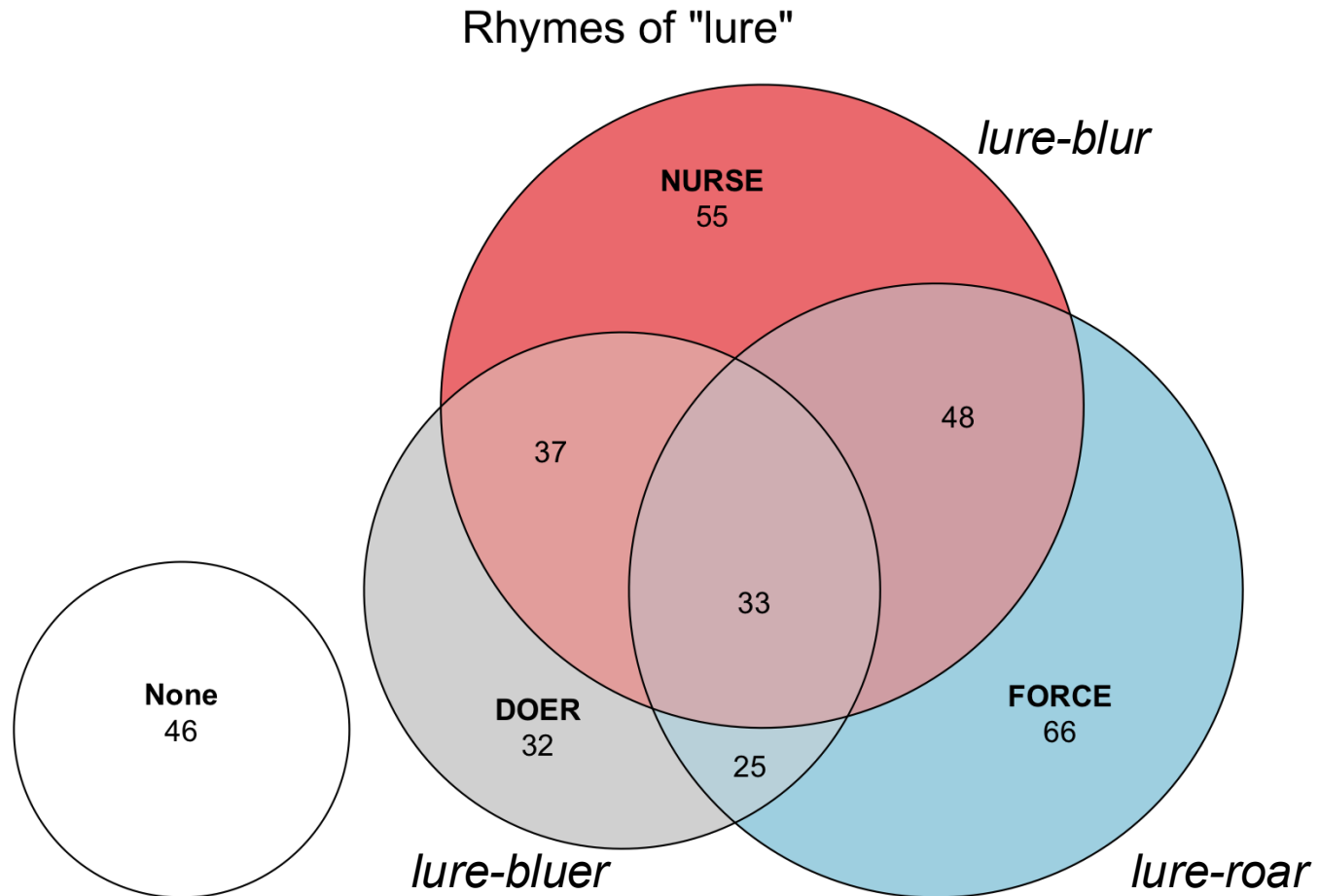
PURE/SURE F2 by item



Interim summary: A conundrum

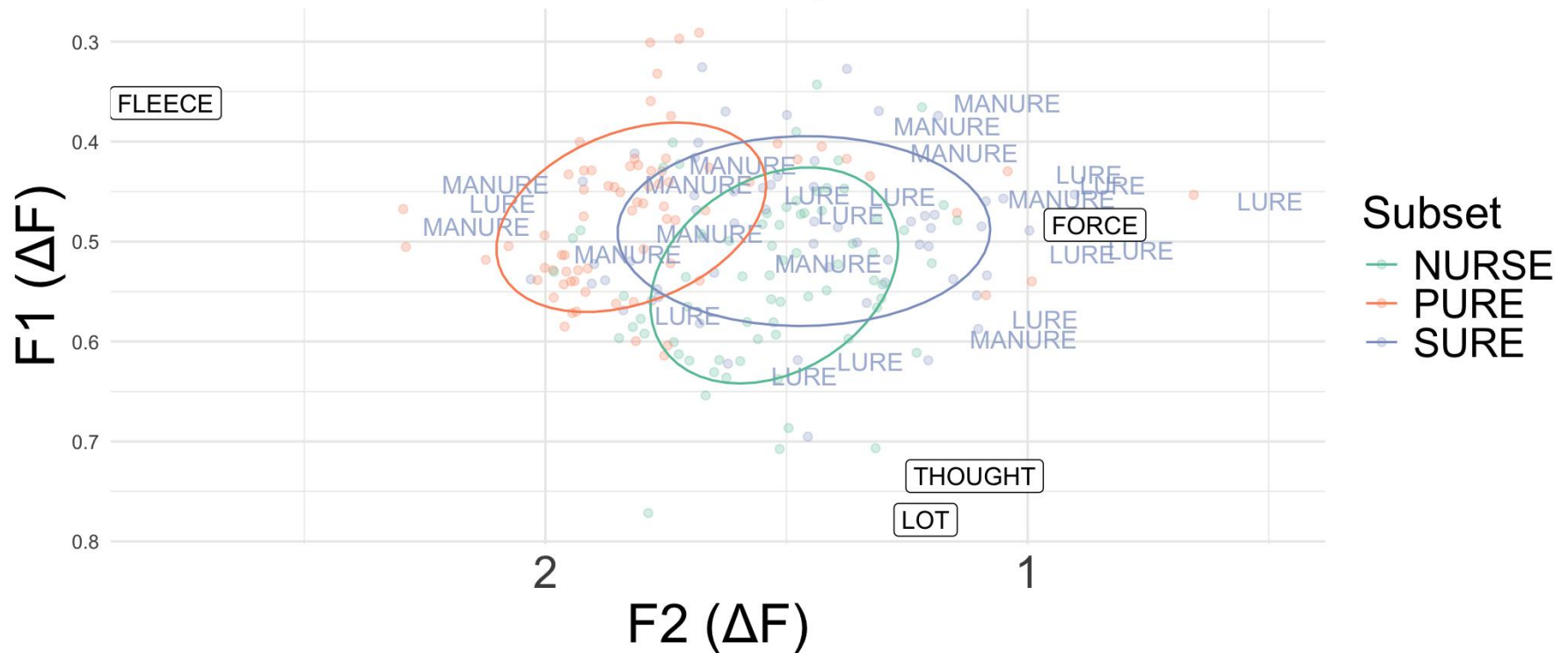
- Merger in perception is advancing in apparent time
- Merger in production appears stable
- Higher PURE-NURSE distance predicts *higher* SURE-NURSE rhyme rate, and vice versa

Lexical items participate in multiple MiPs



MiP does not always translate to MiPr

Formants for speakers who accept
PURE/SURE-NURSE rhymes

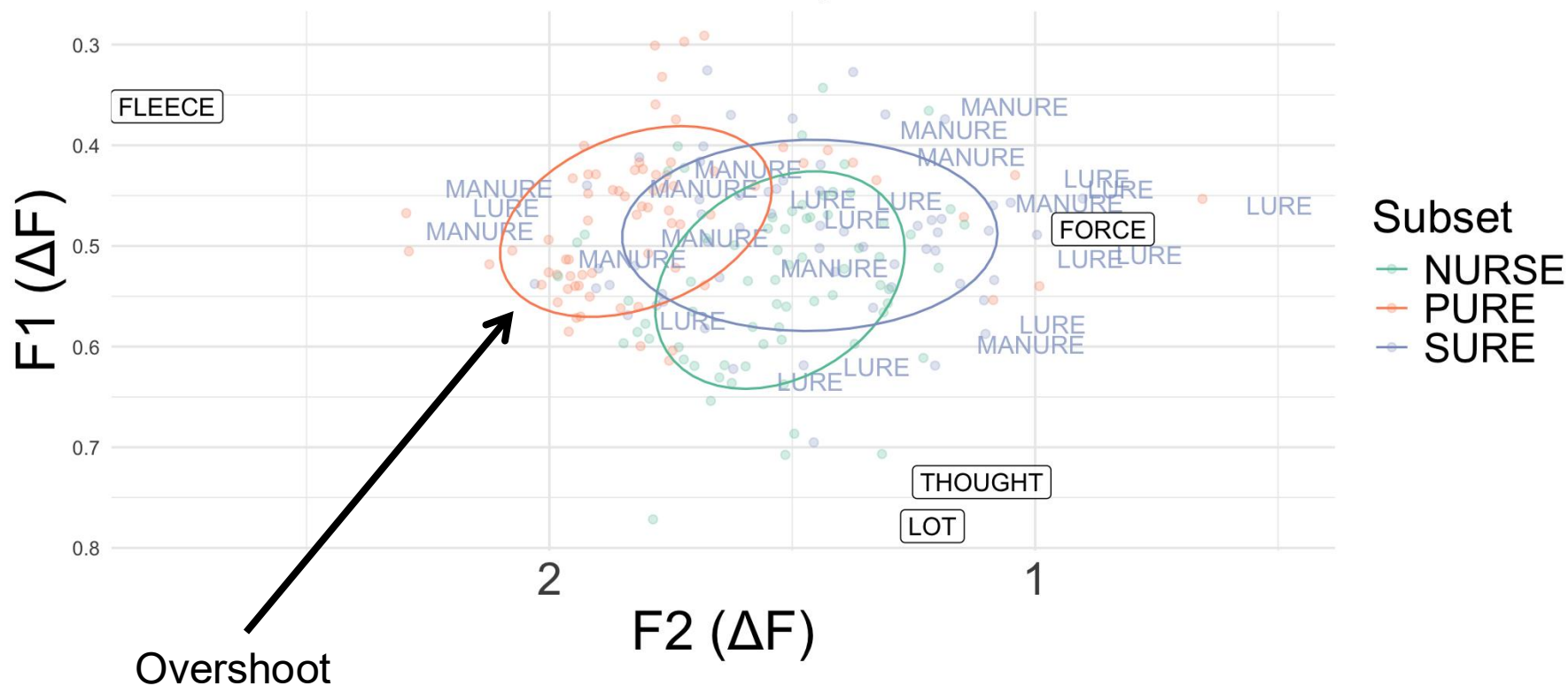


Why does production seem stable?

- Multiple patterns of merger in perception (i.e., with NURSE, FORCE, and DOER) co-exist
- These co-existing mergers in perception translate to variability in production (possibly subject to stylistic factors; see Guenter 2000; Thomas 2008)
- This variability obscures advancing merger in production

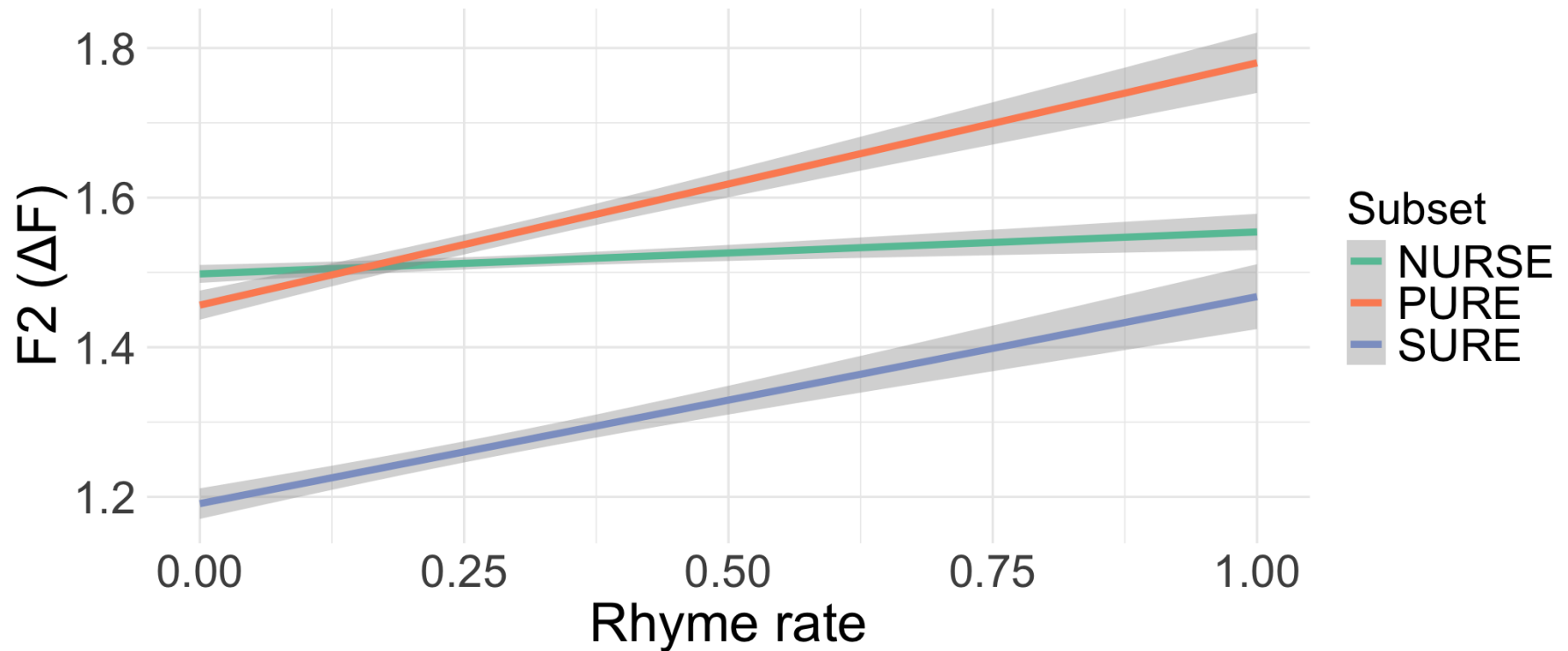
Speakers with MiP have PURE frontier than NURSE

Formants for speakers who accept PURE/SURE-NURSE rhymes



Speakers with MiP have PURE frontier than NURSE

Relationship between PURE/SURE-NURSE rhyme and PURE/SURE/NURSE F2



Why the paradoxical relationship between PURE and SURE?

- Speakers with PURE/SURE-NURSE MiP have PURE fronter than NURSE
- Higher PURE-NURSE distance reflects this “overshoot”
- This creates a correlation between PURE-NURSE distance and SURE-NURSE MiP: speakers with MiP are speakers who front PURE more

Conclusions

1. PURE/SURE-NURSE merger in perception is advancing in apparent time
2. Multiple patterns of merger (i.e., with NURSE, FORCE, and DOER) co-exist and obscure merger in production
3. Merger in perception correlates with merger in production for SURE but systematic “flip-flop” (Hall-Lew 2013) for PURE
 - For merged speakers, this may be interpreted as an allophonic effect conditioned by yod

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Appendix A: Model predicting PURE/SURE-NURSE rhyme judgments

	Estimate	Std. error	z value	Pr(> z)
Intercept	−0.847	0.546	−1.552	0.121
Subset: SURE	−0.933	0.512	−1.821	0.069
PURE-NURSE distance	0.640	1.299	0.493	0.622
SURE-NURSE distance	−2.610	0.868	−3.004	0.003
Year of birth	0.059	0.010	5.976	<0.001
Region: North	1.089	0.420	2.593	0.010
PURE-NURSE distance * Subset	3.298	1.077	3.063	0.002

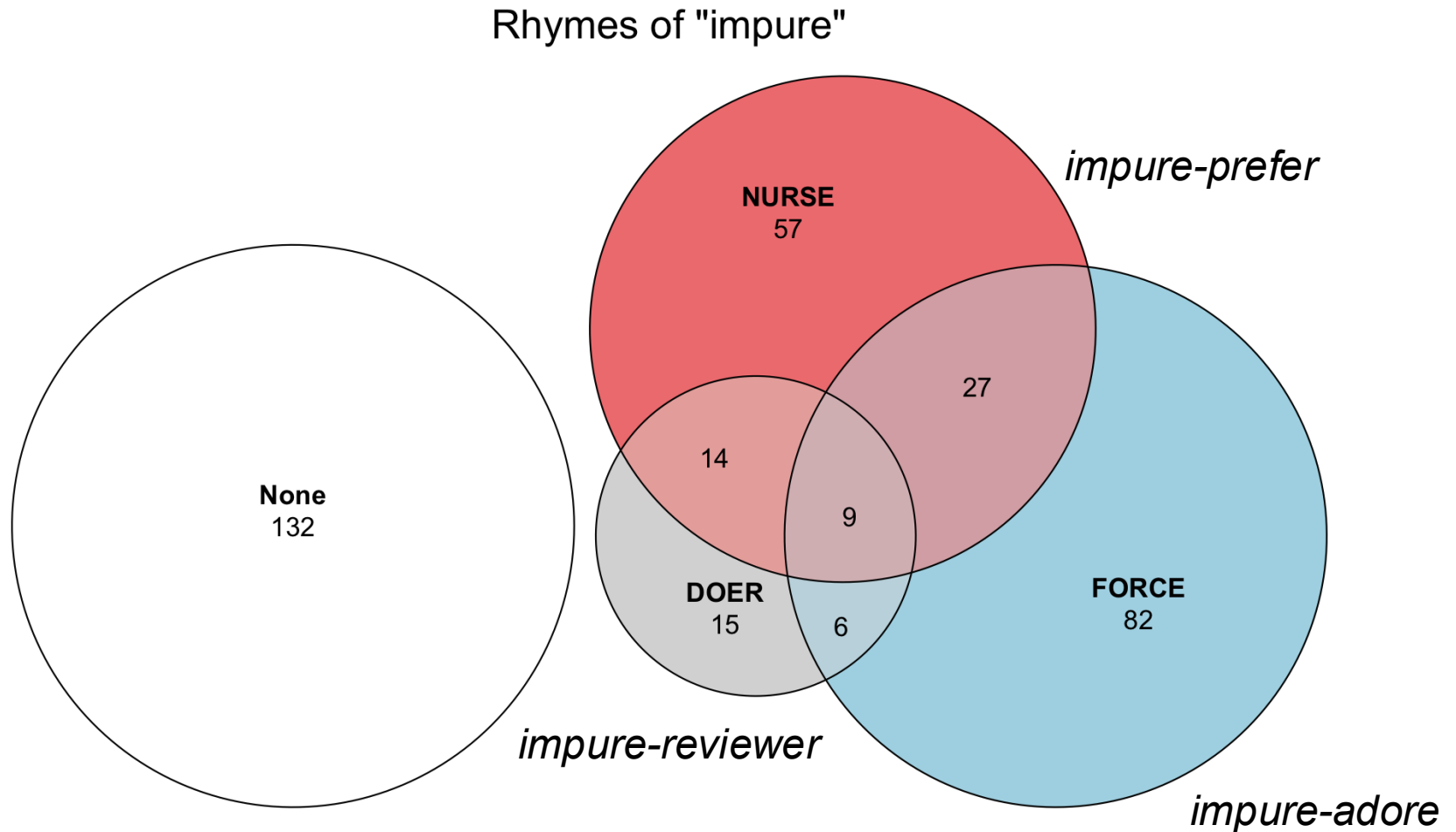
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glmer(Rhyme ~ 1 + Subset + PURE-NURSE distance + SURE-NURSE distance
+ Year of birth + Region + (Subset*PURE-NURSE distance) + (1 |
Participant) + (1 | Word)). N=174. Obs=1737.
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Appendix B: Model predicting distance from NURSE

	Estimate	Std. error	df	t value	Pr(> t)
Intercept	0.296	0.025	13.446	12.086	<0.001
Subset: SURE	0.123	0.032	12.657	3.894	0.002
SURE-NURSE rhyme rate	0.078	0.023	308.470	3.348	<0.001
Region: North	−0.063	0.016	310.095	−3.937	<0.001
Gender: Man	−0.035	0.011	303.748	−3.163	0.002
SURE-NURSE rhyme rate * Subset	−0.148	0.031	306.920	−4.755	<0.001

```
glmer(Distance ~ 1 + Region + Gender (Subset*SURE-NURSE rhyme rate)
+ (1 | Word) + (1 + Subset | Participant). N=313. Obs=3113.
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Appendix C: Euler diagram for *impure*



Appendix D: Euler diagram for *mature*

